



$$x^2 + 4x - 8a \geq 0$$

$$a \leq \frac{x^2 + 4x}{8}$$

$$\cancel{x^4} - \frac{16x^2 + 64a}{(x^2 + 4x - 8a)(x^2 + 4x - 8a)} = \cancel{x^4} + \frac{4x^3 - 8ax^2 + 4x^3}{+16x^2 - 32ax - 8ax^2 - 32ax + 64a^2}$$

$$8x^2 + 32x^2 - 16ax^2 - 64ax =$$

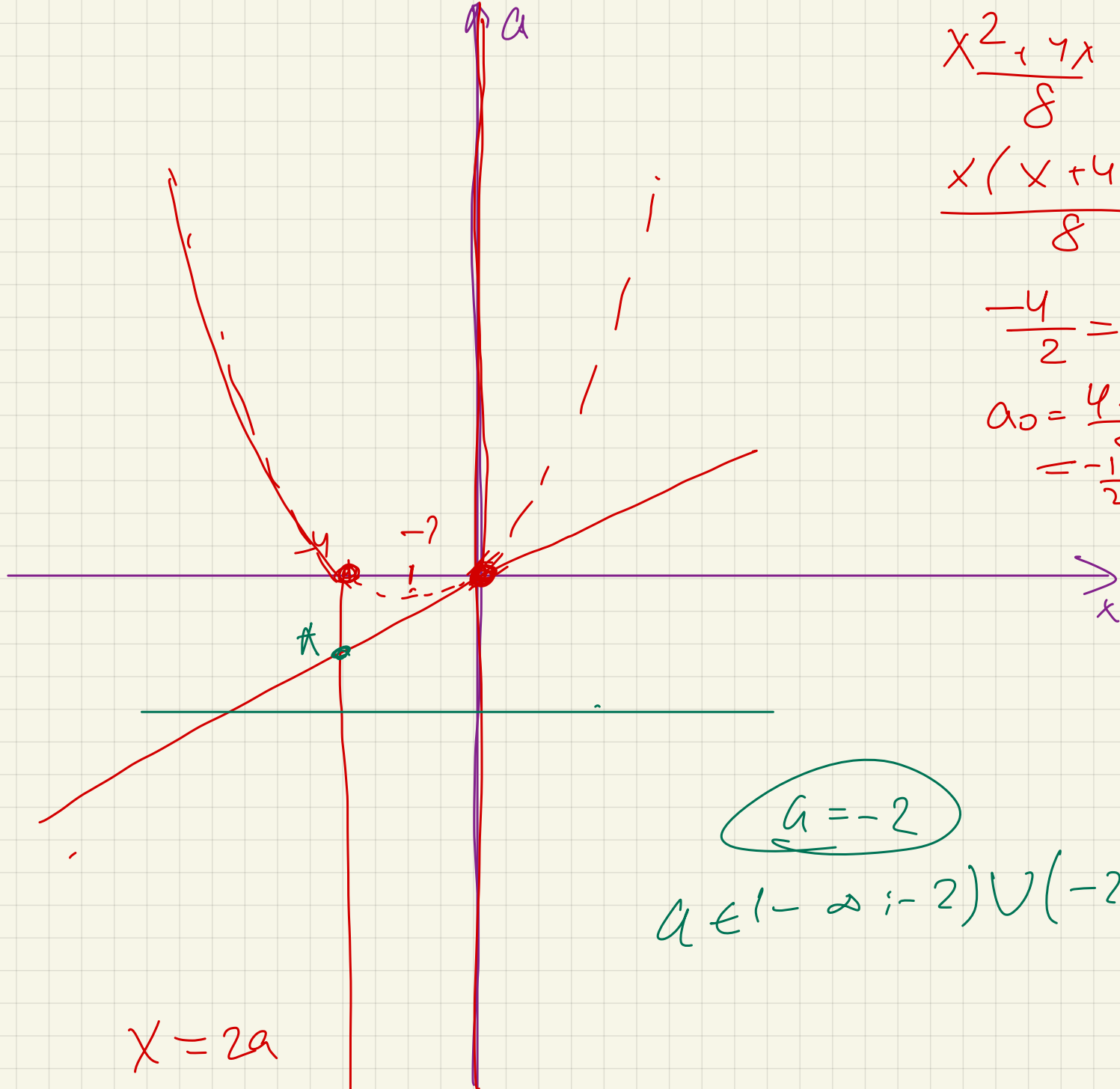
$$8x(x^2 + 4x - 2ax - 8a) = 0$$

$$\left[\begin{array}{l} x = 0 \\ x^2 + 4x - a(2x + 8) = 0 \end{array} \right]$$

$$x(x + 4) - 2a(x + 4) = 0$$

$$x(x + 4)(x - 2a) = 0$$

$$\left[\begin{array}{l} x = 0 \\ x = -4 \\ x = 2a \end{array} \right]$$



$$\frac{x^2 + 4x}{8}$$

$$\frac{x(x+4)}{8}$$

$$-\frac{4}{2} = -2$$

$$a_0 = \frac{4-8}{8}$$

$$= -\frac{1}{2}$$

$$\underline{a = -2}$$

$$a \in (-\infty; -2) \cup (-2; 0)$$

$$x = 2a$$

$$a = \frac{x}{2}$$

$$a = \frac{x^2 + 4x}{8}$$

$$\frac{x}{2} = \frac{x^2 + 4x}{8}$$

$$4x = x^2 + 4x$$

$$x^2 = 0$$

$$500 \text{ z}$$

$$8 \text{ A}$$

$$2 \text{ B}$$

$$200 \text{ z}$$

$$2 \text{ A}$$

$$8 \text{ B}$$

$$1 \text{ A} + 3 \text{ B}$$

$$\begin{array}{c} \text{A} \\ \text{B} \end{array} \quad \begin{array}{c} \text{I} \\ \checkmark \end{array}$$

$$500 - x$$

$$\begin{array}{c} y \\ 200 - y \end{array}$$

$$\begin{array}{l} \text{A: } 8x \\ \text{B: } (500 - x)2 \end{array}$$

$$\begin{array}{c} 2y \\ 8(200 - y) \end{array}$$

$$\sum A = 8x + 2y$$

$$\sum B = 2(500 - x) + 8(200 - y) = 800 + 1600 = 2400 \text{ z.}$$

$$3(8x + 2y) = 2(500 - x) + 8(200 - y)$$

$$S = 8x + 2y$$

$$24x + 6y = 1000 - 2x + 1600 - 8y$$

$$26x = 2600 - 14y$$

$$x = \frac{2600 - 14y}{26}$$

$$S(y) = 8 \left(\frac{2600 - 14y}{26} \right) + 2y$$

$$y = 0$$

$$x = 100$$

$$S = 800 \quad \text{g. A}$$

$$y = a|x - 1| + |a| - 3$$

$$y = \frac{a}{3}$$

$$S \geq \frac{1}{3}$$

$$a = 0$$

$$a > 0$$

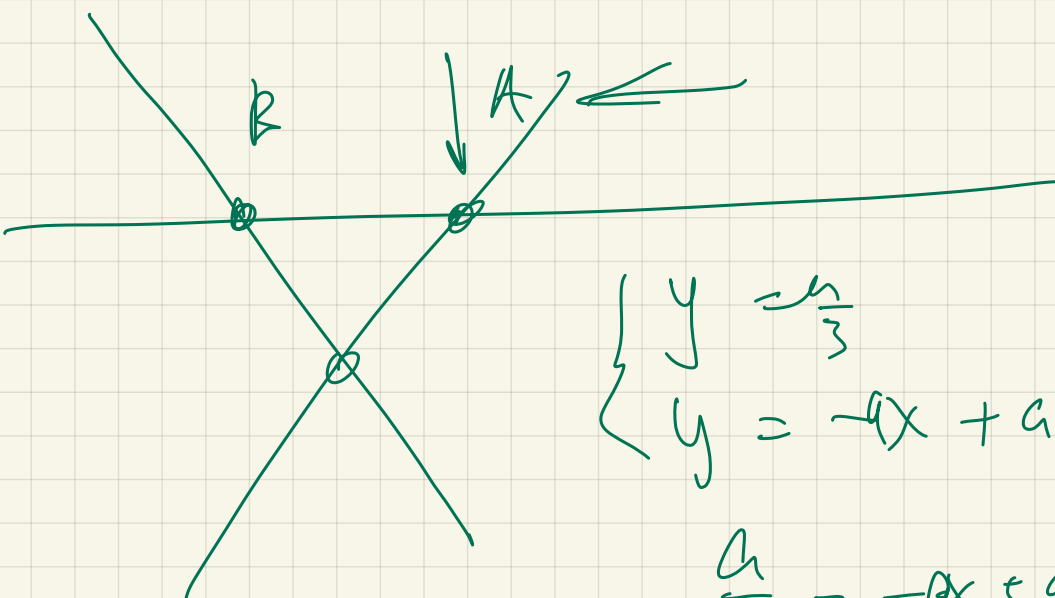
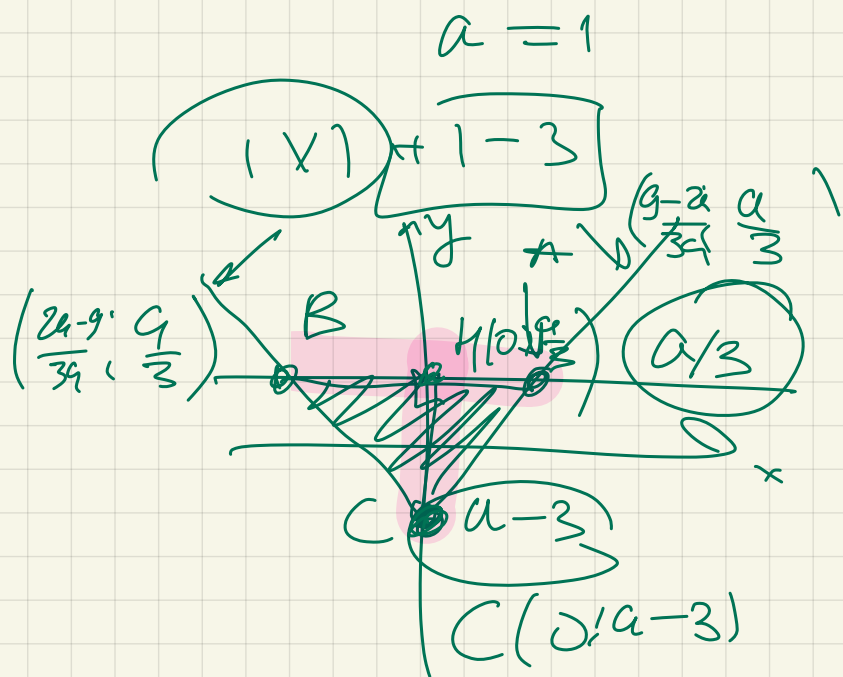
$$a < 0$$

$$y = a|x| + |a| - 3$$

$$a > 0$$

$$a-3 < a \leq \frac{9}{3}$$

$$a < \frac{9}{2}$$



$$\begin{cases} y = \frac{a}{3} \\ y = -4x + a - 3 \end{cases}$$

$$\frac{a}{3} = -4x + a - 3$$

$$4x = \frac{a-3-\frac{a}{3}}{1}$$

$$\frac{2a-9}{34}$$

$$CH = \frac{9-2a}{3}$$

$$AB = \frac{18-4a}{3a}$$

$$\frac{2a-9}{34} - \left(\frac{9-2a}{34} \right)$$

$$a > 0 \quad a < 9/2$$

$$\frac{CA \cdot AB}{2} \geq \frac{1}{3}$$

$$\left(\frac{8-4a}{3a} \right) \left(\frac{9-2a}{3} \right) \geq \frac{2}{3}$$

$$\frac{3(8-4a)(9-2a) - 2 \cdot 3a \cdot 3}{3 \cdot 3a \cdot 3} \geq 0$$

$$\frac{3(72 - 16a - 36a + 8a^2) - 3(6a)}{3 \cdot 3a \cdot 3} \geq 0$$

$$8a^2 - 52a + 72 - 6a$$

$$\frac{8a^2 - 58a + 72}{9a} \geq 0$$

$$f(x) = 3ax^4 - 8x^3 + 3x^2 - 7 \quad [-8; 1]$$

$$a \geq 0$$

1 T. Min.

$$f'(x) = 12ax^3 - 24x^2 + 6x = 0$$

$$6x(2ax^2 - 4x + 1)$$

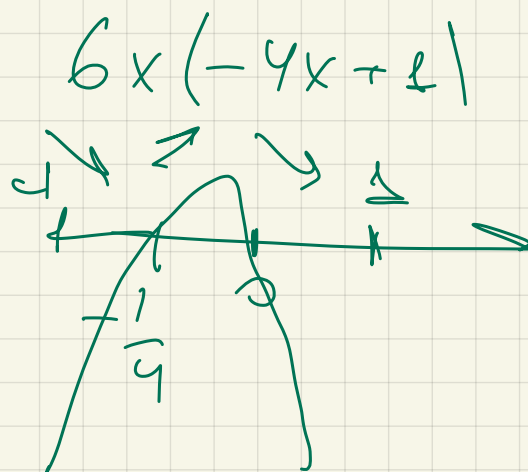
$$\begin{cases} x = 0 \\ 2ax^2 - 4x + 1 = 0 \end{cases}$$

$$4^2 - 4 \cdot a \cdot 2$$

$$4(4 - 2a) \geq 0$$

$$2a \leq 4$$

$$a \leq 2$$

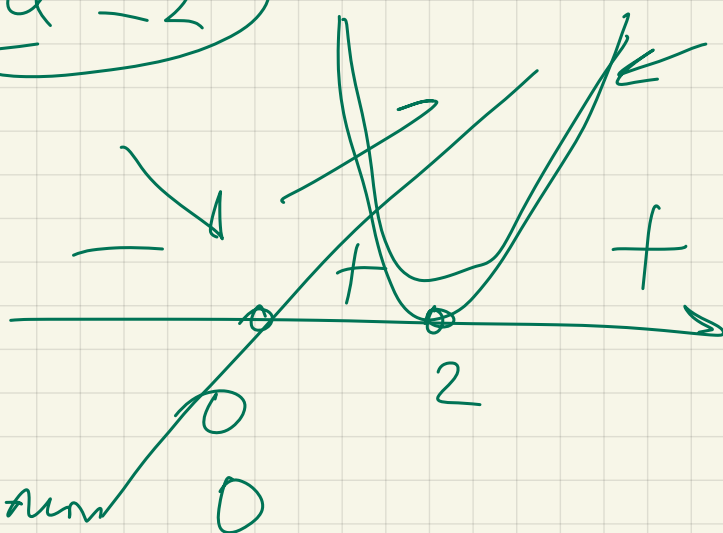


$$a \in [0, 2]$$

$$a > 2$$

$$a = 2$$

1 T. Min.



$$0 < a < 2$$

$$\frac{4 \pm 2\sqrt{4-2a}}{4a}$$

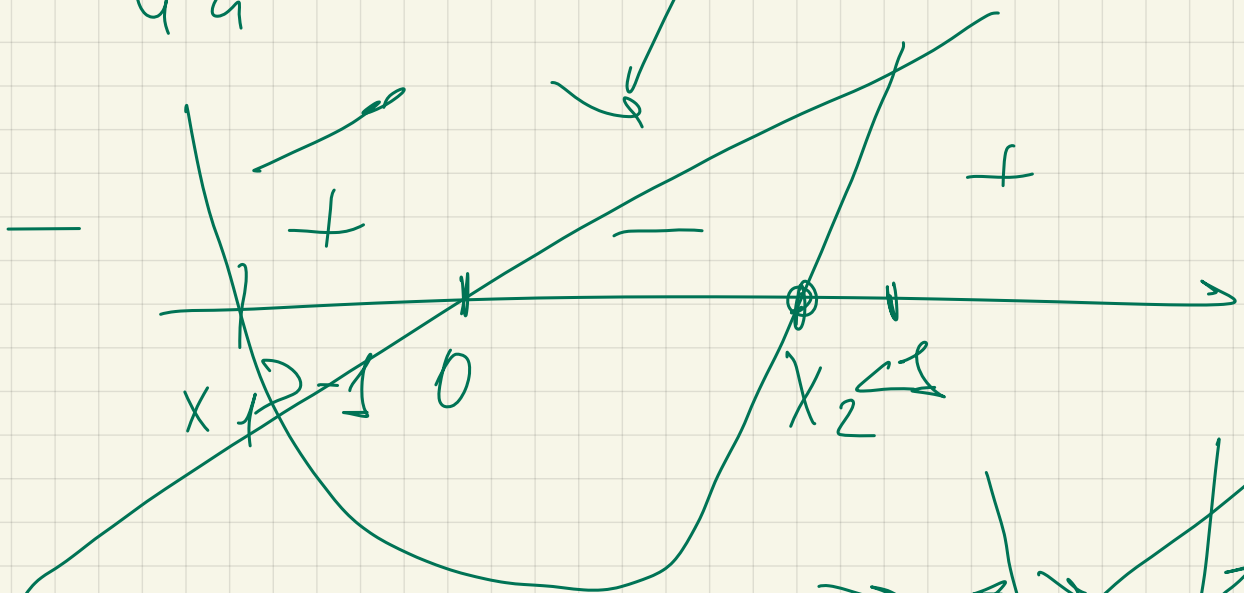
$$-1 \quad 1$$

$$[-1:1]$$

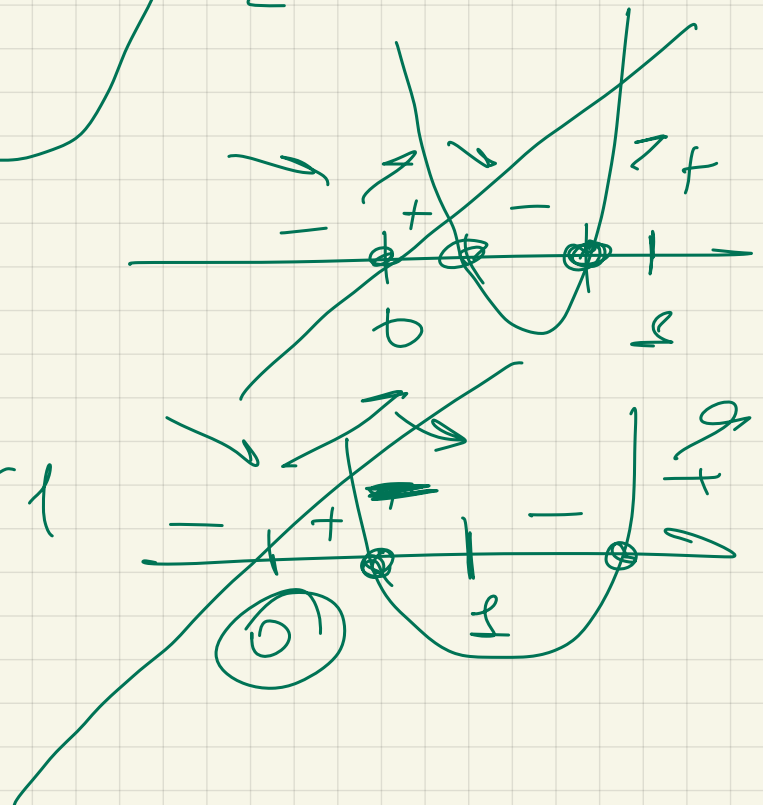
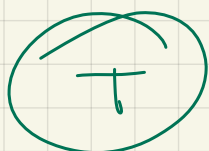
$$\frac{4-2\sqrt{4-2a}}{4a} \quad \frac{4+2\sqrt{4-2a}}{4a}$$

x_1 x_2

$$\frac{4+2\sqrt{4-2a}}{4a} < -1$$



$$\left\{ \begin{array}{l} \frac{4-2\sqrt{4-2a}}{4a} < -1 \\ \frac{4+2\sqrt{4-2a}}{4a} < 1 \end{array} \right.$$



$$0 < x_1 < 1 \quad \checkmark$$

$$x_2 > 1$$

$$0 < \frac{4 - 2\sqrt{4-2a}}{4a} < 1$$

$$4 - 2\sqrt{4-2a} < 4a$$

$$4 - 4a < 2\sqrt{4-2a}$$

$$(4 - 4a)^2 < 4(4 - 2a)$$

$$\rightarrow \textcircled{+}$$

$$a = 0$$

$$a \in [0; 7] \cup]7; +\infty)$$